

CS 6762
Fall 2023
SP, ML, FC Homework 3

Answer all questions. Work alone. Total is 100 points. Type or write neatly.

1. Describe entropy, information gain and the relationship between the two. (5 points)
2. Assume an urn has 3 red balls, 2 green balls, and 1 white ball. Compute the entropy. (5 points)
3. What is a random forest classifier and state its advantages over a decision tree. (5 points)
4. What is a sigmoid function? A tanh function? A softmax? (5 points)
5. Describe in 1 or 2 paragraphs how to train a NN (look this up on the Internet and/or watch some YouTube videos). (10 points)
6. Briefly describe 2 applications from any of the smart health, smart cities, or autonomous systems application areas where you can/should use a DNN. (10 points)
7. CNNs were originally developed for use with images. Explain how to use CNNs for 2 problems that are NOT related to images. For this, explain the inputs and eventual outputs of the CNN. (20 points)
8. Describe the main gates in an LSTM cell, including their purpose, their inputs and outputs, and what is inside the gate. (10 points)
9. What are the key issues in developing DNNs for execution on small devices? (5 points)
- ~~10. In the SparseSep system we studied, what are the inputs and outputs to its Layer Compression Compiler? (5 points)~~
11. Describe how ML can be used in autonomous systems and what are its challenges? (10 points)
12. Describe how ML can be used in smart health CPS systems and what are its challenges? (10 points)

Note: Questions 11 and 12 are not the same as question 7 which is more related to the mechanics of the CNN, although there may be some overlap.